Group 6: BSMM 8730: D.4.2. Project Documentation: Design

Why Are Patients Waiting So Long to Get New Kidneys in Canada?

1. Introduction and Context

This project will develop an interactive Power BI dashboard to examine publicly reported Canadian kidney-transplant indicators, including donor activity, transplant volumes, wait-list measures, and reported outcomes. Selected United States indicators may be included to provide context where definitions, populations, coverage, units, and reporting periods are sufficiently compatible.

The proposed workflow uses Python to acquire and prepare approved public data, MySQL to store validated records, and Power BI to present interactive results. The intended audience includes healthcare administrators and policy planners, organ-donation organizations and awareness teams, and members of the public seeking an evidence-based overview.

1.1 Scope

Included	Excluded
Acquisition of approved public datasets; Data cleaning and standardization; Relational storage; Documented indicator calculations; Dashboard development; Validation; Source and methodology documentation; Selected policy context where reliable and relevant.	Private patient records or personally identifiable health information; Clinical advice or decisions; Organ allocation; Individual transplant predictions; Causal claims about why delays occur; Guaranteed coverage of every province, territory, or U.S. state.

A key interpretation rule is that dialysis-to-transplant time is not interchangeable with formal time on a transplant wait-list. Each measure will retain its published definition, population, unit, reporting period, and limitations. The dashboard will describe observed patterns, not infer causation.

1.2 Design Alignment and Traceability

Requirement area	Design response
FR-01 to FR-06	Source-specific acquisition, provenance capture, staging, validation, approved database loading, and reporting data preparation. The exact API/JSON access method required by FR-01 must be confirmed for each source.
FR-07 to FR-11	Indicator presentation, filtering, comparison, and explanatory context will follow the finalized requirement wording and associated acceptance criteria. Comparisons are gated by documented compatibility checks.
FR-12 to FR-14	Reset behaviour, export where supported/permitted, and reviewed policy context are treated according to their approved priority and acceptance criteria. Optional features are not assumed to be mandatory.
NFR-01 to NFR-10	Data quality, traceability, performance, security, reproducibility, and supported-environment constraints are addressed throughout the design.
AC-01 to AC-14	Acceptance tests will be recorded by requirement ID, expected result, actual result, status, and reviewer. AC-05 requires independently calculated values to agree when the same source data and definitions are used.

The final traceability matrix will be checked against the team's-controlled user requirements document (D.4.1) before submission, so each individual FR, NFR, and AC is mapped to the exact approved wording.

2. Goals and Non-Goals

The goal is to provide accurate, understandable, and traceable information about available kidney-transplant indicators. The targets below are proposed acceptance criteria for the project, not results already achieved.

Goal	Measurable success criterion	Related requirement
Data quality	All approved records pass required-field, type, duplicate, value-range, and relationship checks, or are explicitly quarantined with an issue record.	NFR-01; NFR-06
Calculation accuracy	Selected dashboard values are independently recalculated from the same source records and definitions and reconciled in the test log.	NFR-02, AC-05
Usable interaction	Year and geography filters update all applicable visuals; reset restores documented defaults.	FR-08; AC-06; FR-12
Comparability	Every proposed Canada- U.S. comparison has a recorded compatible, qualified, or incompatible decision and an explanation.	FR-10; AC-08
Traceability	Each displayed indicator can be traced to its source, reporting period, definition, unit, and calculation rule.	NFR-07, NFR-09
Performance	Standard dashboard pages meet the approved 10-second target under the agreed normal test conditions, excluding initial refresh and external network delays as specified in the requirements.	NFR-03
Reproducibility	A second team member can follow the documented steps to rebuild the database and refresh the report using the approved project files and environment.	NFR-09

2.1 Non-goals

- Real-time reporting or continuous clinical monitoring.
- Clinical recommendations, individual risk scoring, or predictions about a patient's transplant timing.
- A complete census of every Canadian jurisdiction or U.S. state regardless of source availability.
- Direct comparison of measures whose definitions, populations, or denominators are not sufficiently aligned.

3. System Architecture

The system uses a controlled batch workflow. Public source material is acquired, staged, standardized, and validated before approved records are loaded into MySQL. Power BI reads approved reporting tables or views. Users interact with the report and do not edit the underlying database.

High Level System Architecture

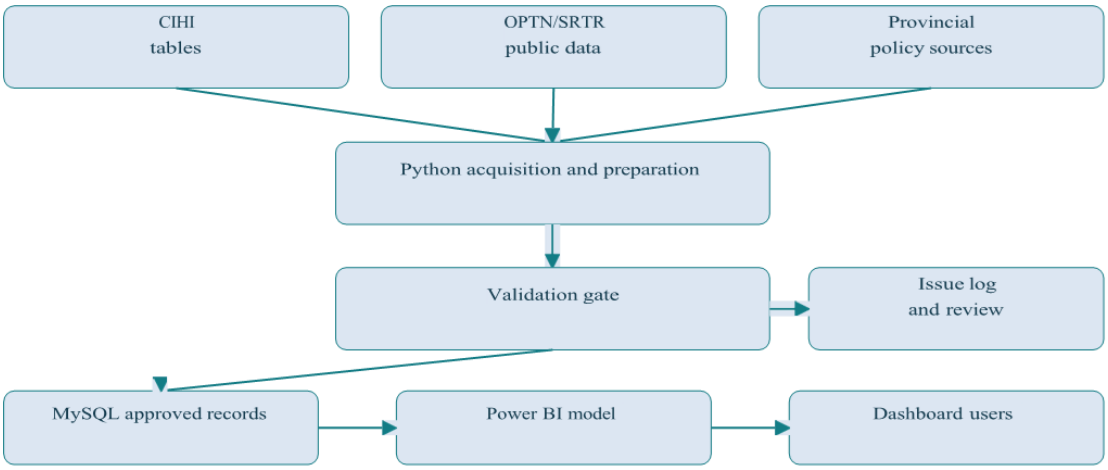


Figure 1. Proposed high-level system structure. Failed checks trigger review before records are approved.

3.1 Main subsystems

Subsystem	Responsibility	Input → Output
Source acquisition	Retrieve approved public files, accessible tables, or confirmed API responses; preserve provenance and original inputs.	Published source → raw/staged records
Preparation and validation	Map fields, standardize periods/geographies/categories/units, check quality, and record exceptions.	Staged records → approved or quarantined records
Database	Store validated observations and supporting metadata with keys and constraints.	Approved records → relational tables/views
Analytics	Apply indicator calculations, aggregation rules, and Canada-U.S. compatibility decisions.	Validated records → reporting measures
Presentation	Display interactive pages, filters, definitions, sources, and data limitations.	Reporting views → Power BI dashboard

4. Logical Architecture

The logical design separates source-specific retrieval from shared transformation, validation, analytics, and presentation functions. This reduces the impact of a source-format change and makes quality controls reusable.



Figure 2. Logical architecture and separation of responsibilities.

Layer	Core functions
Acquisition	Source adapters; approved file/API retrieval; retrieval date, URL, source title, version, and original filename capture.
Staging and mapping	Preserve raw inputs; map source columns to the project data dictionary; standardize geography, category, period, and units.
Quality controls	Required-field, type, range, duplicate, referential-integrity, and reconciliation checks; issue logging; approval or quarantine.
Analytics	Indicator calculations; aggregation rules; source-selection rules; comparison compatibility decisions; reporting views.
Presentation	Power BI pages, slicers, reset behaviour, explanatory notes, and source/methodology information.

Policy documents are reviewed and coded separately from numerical observations. Policy context may help describe the environment but will not be treated as evidence that a particular policy caused a measured outcome.

5. Physical Architecture

The initial implementation is intended for a team-managed environment. Exact operating systems, hardware, connector versions, database host details, and deployment permissions must be recorded by the team before performance testing.

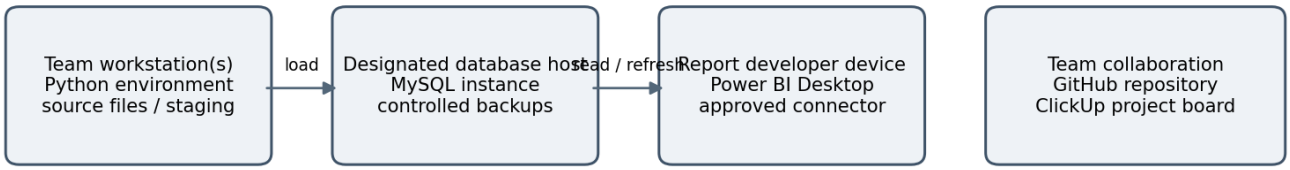


Figure 3. Proposed physical arrangement. Actual host names and infrastructure remain to be confirmed.

Element	Proposed arrangement	Deployment condition
Python environment	Team workstation(s) with documented Python version, dependencies, and reproducible scripts.	Record actual version and dependency list; avoid assuming a specific minor version before setup.
MySQL	Designated team database host for integration, constraints, reporting views, and controlled exports.	Confirm host, version, access method, backup location, and team permissions.
Power BI	Power BI Desktop report connected to approved MySQL tables/views using an available connector.	Power BI Service publication and scheduled refresh are conditional on licensing, access, gateway, and institutional permissions.
GitHub /ClickUp *(Collaboration)*	GitHub for code/document version control; ClickUp for tasks, issues, and acceptance-test tracking.	Do not commit credentials, private data, or unapproved source extracts.

The project will record the actual operating system, software and connector versions, available memory, database location, and dataset size. Hardware requirements such as a fixed minimum RAM specification are not asserted until the team has tested its actual workload.

6. Data Model

6.1 Design principles and grain

The core model is observation-based rather than a single fixed fact table containing every metric. One Observation represents one published value for a defined indicator, geography, reporting period,

category combination, and source dataset. This accommodates different units, populations, and reporting grains without implying that unlike measures can be aggregated together.

A separate indicator definition records the measure's name, definition, population, unit, and aggregation rule. The model will be refined after inspecting representative source extracts. Fields and categories that are not present in the source data will not be assumed.

Entity relationships

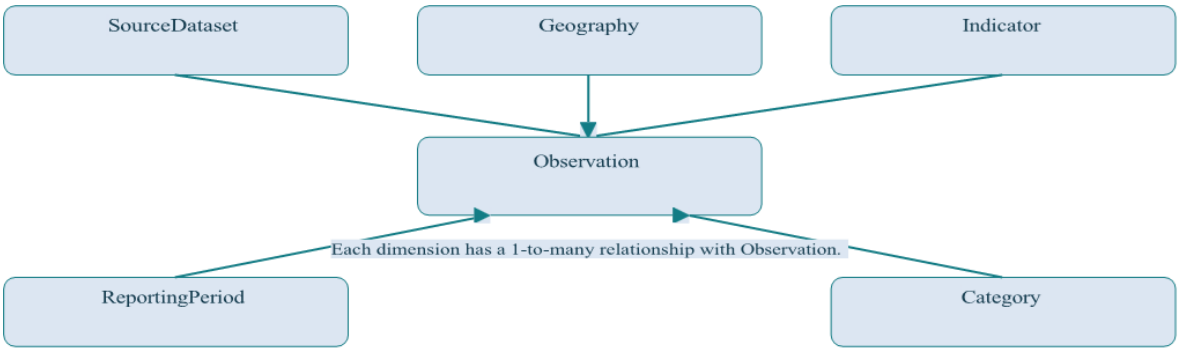


Figure 4. Conceptual entity relationship model. Final keys, optionality, and cardinalities must be validated against actual source extracts.

Entity	Key fields	Purpose
SourceDataset	dataset_id (PK), organization, title, URL, retrieval_date, version.	Records provenance for each source extract
Geography	geography_id (PK), geography_name, geography_level, country_code.	Supports province/territory, national, and selected U.S. geographies.
Reporting Period	period_id (PK), start_date, end_date, period_label, period_type.	Distinguishes annual periods from point-in-time snapshots.
Indicator	indicator_id (PK), name, definition, population, unit, aggregation_rule.	Defines the meaning and valid handling of each measure.
Category	category_id (PK), category dimensions and source-defined labels.	Stores only categories supported by the source.
Observation	observation_id (PK), dataset_id, geography_id, period_id, indicator_id, category_id (FKs), value, value_status, notes.	Stores the published value and its status.
PolicyContext	context_id (PK), dataset_id and geography_id (FKs), topic, summary, review_date, source_reference.	Stores reviewed contextual notes separately from numeric observations.
ComparisonRule	comparison_rule_id (PK), indicator pair, decision, rationale, review_date.	Documents whether a proposed cross-border comparison is compatible, qualified, or incompatible.

6.2 Integrity and aggregation rules

- Required foreign keys must reference valid dimension records. A uniqueness constraint will prevent duplicate observations at the agreed observation grain.
- Missing, unavailable, and source-suppressed values remain NULL with an explicit value_status; they are not converted to zero.
- Counts must be non-negative. Units and denominators must be retained for rates and normalized measures.
- National totals must not be added to their provincial components. Provincial medians must not be averaged to create a national median.
- Donor counts, kidney transplant counts, wait-list snapshots, and annual activity measures remain distinct indicators unless the source definition explicitly establishes equivalence.
- Overlapping source records may be retained for provenance, but reporting views must select the approved source to prevent double counting.

A data dictionary will define field types, allowed values, category mappings, units, null/status handling, and aggregation rules. The final ERD should be updated after the first representative source extracts are inspected.

7. Component Design

7.1 Acquisition, preparation, and validation

The acquisition module will use the method permitted and available for each approved source. API/JSON retrieval will be used where a suitable endpoint and access method are confirmed. If a source is available only as a published file or accessible table, the team must document the method and resolve any conflict with FR-01 through the approved requirements-change process.

Pseudocode: batch ingestion and quality gate

```
FOR each approved source:
    Retrieve source using its confirmed access method
    Save original input without altering it
    Record URL, title, retrieval date, version and filename
    Extract records into staging
    Map source fields to the project data dictionary
    Standardize geography, category, period and unit
    Preserve missing and suppressed-value status
    Run required-field, type, range, duplicate and FK checks

    IF a critical check fails:
        Write issue details to the validation log
        Quarantine affected records for review
    ELSE:
        Load approved records in a database transaction
        Reconcile row counts and selected source values
        COMMIT only when reconciliation succeeds
        Otherwise ROLLBACK and log the failure
```

7.2 Example database validation

The following SQL illustrates a referential-integrity check for observations whose geography key has no matching dimension record. The final table and column names must match the implemented schema.

```
SELECT o.observation_id, o.geography_id
FROM Observation AS o
LEFT JOIN Geography AS g
```

```
  ON o.geography_id = g.geography_id
WHERE g.geography_id IS NULL;
```

The production database should also enforce foreign keys directly. The query is a diagnostic check, not a substitute for database constraints.

7.3 Indicator and comparison logic

Year-over-year percentage change is calculated only when the current and previous observations use the same indicator definition, population, unit, geography, category, and compatible reporting periods, and the previous value is greater than zero:

((current value − previous value) / previous value) × 100

If the prior value is zero, missing, suppressed, or not comparable, the dashboard will show an explanatory unavailable-data message rather than a misleading percentage.

Compatibility check	Decision rule
Definition and population	Confirm that both measures represent the same concept and target population.
Unit and denominator	Confirm identical units or a valid documented conversion, including compatible denominators for rates.
Period and coverage	Confirm aligned reporting periods, geographic coverage, and inclusion criteria.
Outcome	Compatible: direct comparison allowed. Qualified: display only with a visible caveat. Incompatible: do not compare directly.

7.4 Refresh and acceptance workflow

- Data engineer reviews source changes, runs ingestion and validation, and approves or quarantines the update.
- Database records are loaded transactionally and reconciled against source row counts and selected published values.
- Dashboard developer refreshes Power BI and checks filters, labels, period coverage, missing-data messages, and selected totals.
- Analyst independently recalculates selected indicators and records the result against AC-05.
- Team records each test with requirement ID, input/action, expected result, actual result, Pass/Fail/Not Applicable, reviewer, and date.

8. Interface Design, Constraints and Security

8.1 Dashboard structure

Page	Proposed content
Overview	Selected indicator cards, Canadian trend summaries, active filters, coverage notes, and links to detailed pages.
Provincial Patterns	Available provincial/territorial indicators over time, with

	geography and period context.
Donors and Wait-Lists	Donor activity, transplant volumes, wait-list counts, and reported wait-list outcomes, kept as distinct measures.
Canada-U.S. Context	Only comparisons assessed as compatible or qualified, with visible definitions and caveats.
Sources and Methods	Source names, reporting periods, indicator definitions, calculations, missing-data explanations, and reviewed policy context.

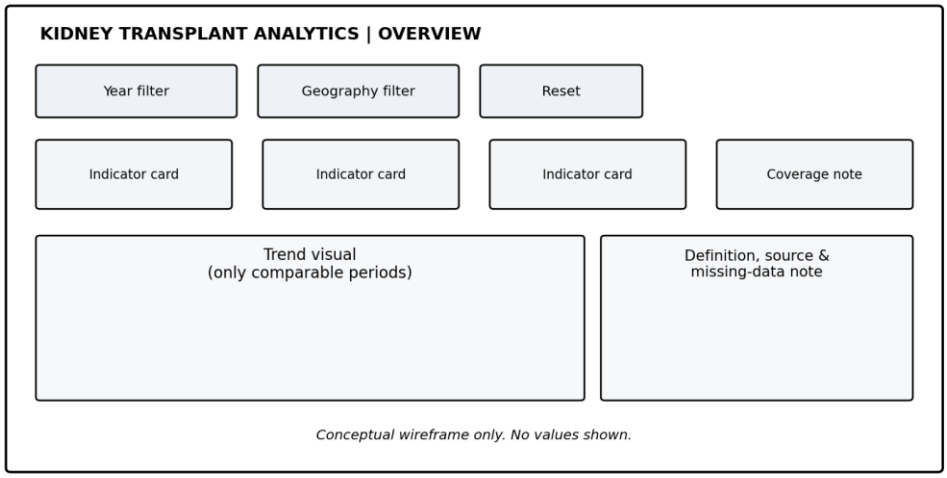


Figure 5. Conceptual overview-page wireframe. It contains no actual or assumed data values.

Filters will affect only the visuals to which they apply. If a selected year or geography has no observation, the report will explain that the value is unavailable rather than silently substituting another period. Each visual will identify its reporting period when periods differ. Colour will not be the only way to communicate a category. Reset will restore documented defaults.

Export will be enabled only where supported by the selected Power BI environment, permitted by source terms, and consistent with the finalized requirement priority. It is not treated as guaranteed capability.

8.2 Data interfaces and protocols

Connection	**Design**
Public source → Python	HTTPS retrieval of approved published files or confirmed API responses. Source-specific access method and format are documented.
Python → MySQL	Parameterized database operations using the approved connector; transactional loads; separate staging and approved data.
MySQL → Power BI	Power BI reads approved tables or reporting views through an available

	MySQL connector. Refresh is batch-based.
Team collaboration	Git-based version control for code and documentation; ClickUp for task and issue tracking.

No custom public-facing API is proposed. The system is designed around periodic published data, not live clinical feeds.

8.3 Constraints and security

- Source formats, reporting frequency, incomplete geographic coverage, suppressed values, and changes to published definitions may constrain the analysis.
- Power BI performance depends on dataset size, query design, model relationships, visuals, available hardware, and connector behaviour. The agreed 10-second target will be tested in the documented environment.
- Only public aggregate information will be collected. Patient identifiers will not be intentionally stored or displayed, and suppressed values will not be reconstructed.
- Credentials and connection strings will be kept outside source code and version control. Access will use separate write and reporting-read permissions where feasible.
- Database exports and approved project files will have recoverable copies. Source licensing, terms of use, and attribution requirements will be followed.
- Policy text will be reviewed for relevance and accurately attributed; it will not be used to infer causation without suitable evidence.

9. Design Rationale

Design choice	Rationale
Python for acquisition and preparation	Supports repeatable file/API handling, data cleaning, validation, structured logging, and text processing.
MySQL relational storage	Provides keys, constraints, structured relationships, repeatable queries, and reporting views.
Observation-based data model	Accommodates indicators with different units, populations, categories, and reporting periods without forcing unlike measures into one fixed fact structure.
Power BI with batch refresh	Supports interactive reporting from periodically published aggregate data without implying real-time source availability.
Separate policy context	Preserves the distinction between descriptive numerical observations and contextual material that may not establish causal relationships.

Compatibility review before comparison	Reduces the risk of presenting superficially similar but methodologically different Canadian and U.S. indicators as directly comparable.
GitHub and ClickUp	Support collaboration, reproducibility, change tracking, task ownership, and issue resolution.

The overall design prioritizes accuracy, interpretation, and traceability over maximizing the number of displayed indicators. Where the source material cannot support a calculation or comparison, the dashboard will explain the limitation and restrict the affected analysis.

10. Source Basis and Revision History

The design draws on the project's Approved Proposal, the User Requirements and Analysis Document, and the technical and functional considerations established during the design process. The proposed solution also references CIHI and OPTN/SRTR as its primary data sources. As implementation progresses, the team will maintain a record of the datasets used, including their titles, source URLs, reporting periods, retrieval dates, and relevant methodological notes. Any data preparation or transformation applied will also be documented to support transparency and reproducibility.

Version	Date	Change	Status
1.0	Sep 27, 2026	Initial project proposal submitted.	Returned for Revision
1.1	October 2, 2026	Revised proposal to address data integration, web scraping, and MySQL implementation.	Resubmitted
1.1	October 4, 2026	Revised proposal reviewed; team authorized to proceed.	Approved to proceed
1.0	October 23, 2026	User Requirements and Analysis Document submitted.	Awaiting feedback
1.0	October 30, 2026	Design document	Awaiting feedback